Feature selection | filter method

Feature $\longrightarrow$ input column

p input feature $\longrightarrow$ subset (s) of p is used.

$p > s$.

Like 10 columns but we use 3 columns.

This is feature selection.

Why use feature selection?

1. Curse of Dimensionality : With certain no. of feature we get optimum result.
2. Computational Complexity
3. Interpretability : Explaining the inference (relation) is easy.

Types of Feature Selection :

Filter Based | Wrapper method | Embedded techniques | Hybrid techniques

Filter Based Technique : It uses statistical methods

$f_1, f_2, f_3, \ldots f_n$.

Each feature is studied statistically & decided whether to keep it or remove it.

- Variance Threashold.
- Correlation based filteration.
- ANOVA
- Chi-square.
- Mutual information.

Duplicate Features :

Removing the duplicate features.

```
df = df.loc[:, ~df.T.duplicated()]
```

Variance Threshold :

It's applied on two types of features :

1. Constant features

A	B	C
1	1	Y
2	1	N
3	1	X

↳ Constant Column

Variance = 0

var = 0 .

∴ Drop such column.

2. Quasi- constant columns.

	A	B	C
$n = 1000$	–	–	–
	–	–	–
	–	–	–

↳ 995 rows value = 1

5 rows value = 0

∴ Quasi- constant.

Drop such columns.

Step 1 : Threshold = 0.1

Step 2 : f_1 ⟶ variance. for all features.

Step 3 : var(f) < Threshold : Drop those columns.

Keep threshold b/w 0.01 to 0.1.

usually = 0.05.

Points to consider :

1. Ignores target variables → We do not study relation b/w x_i & y ; we just study var(x_i).
   It's a univariate method. Possibility of keeping irrelevant feature with high var.
   Similarly, if x_i is very relevant & have low var. we may drop it.

2. Ignores feature interaction ⟶ f1 and f2 may have relation.
   eg) f1 - longitude → var ↑ → if it's added mistakenly.
   f2 - latitude → var ↓

3. Sensitive to feature scaling ⟶ eg) f_1 - values in lakh.
   f_2 - value in 0.2
   & we took threshold = 0.1, then
   f1 will never be dropped.
   Scaling is necessary.

4. Arbitrary Threshold Value → Challenging to determine the threshold value.
   Generally b/w 0.01 to 0.1.

Correlation : Pearson corr. cofficient.

i/p col. o/p col

Pick → f_1 y

Trying to find the linear relation b/w f1 & y.

Number generally b/w -1 to 1

Strong inverse L.R Strong +ve L.R.

eg) coff = 0.9, then if x↑, y↑

$f_1, f_2, \ldots f_n, y$

corr (f_1, y)

corr (f_2, y)

⋮

corr (f_n, y)

Approach 1

Set a cutoff corr. coff → Suppose 0.3.

If corr $(f_1, y) > 0.3$: if true keep else drop.

* Approach 2

corr (f_1, f_2)

corr (f_2, f_3)

then look for multicollinearity.

eg) corr $(f_1, f_2) = 0.9$

corr $(f_1, f_3) = 0.85$

Drop

Disadvantages :

1. Linearity Assumption → May be x_1 has strong non-linear relation with y.
2. Doesn't capture complex relationships → if (f_1, f_2) have linear relation, but (f_1, f_2, f_3) may be very useful to predict y.
3. Threshold determination
4. Sensitive to outliers. → Remove outliers.

ANOVA :

Hypothesis Test
↓

	x_1	Y	
In this lecture →	Numerical	Categorical (Classes > 2)	[1-way ANOVA]
OR	Numerical	Numerical	

In our Dataset ↓

$\boxed{f_1, f_2, \ldots f_{153}}$ $\boxed{y_{\text{category} = 6 > 2}}$

$f_1 \rightarrow y$: if relation = Strong, keep else drop.

Calculate F-statistic $= \dfrac{MS \text{ between}}{MS \text{ within}}$

MS ↓ mean sq.

(* Look in hypothesis test notes)

Disadvantages :

1. Assumption of Normality.
2. Assumption of Homogenity of variance.
3. Independence of observation.
4. Effect of outliers.
5. Doesn't Account for Interaction.

Chi-Square Test :

Tells relationship b/w f_1, y

$f_1 \downarrow$ Cat, $y \downarrow$ categorical.

Both should be categorical column.

Is there relation b/w sex & survival, if True keep else drop.

Contingency table ← observed value. [Real data]

Survived → (columns), Sex → (rows)

	0	1	
M	468	109	ΣM
F	81	238	ΣF
	Σ0	Σ1	Total

Expected value [ideal data]

	0	1
M	$\frac{\Sigma M \times \Sigma 0}{Total}$	355
F	120	221

Observed $\xleftrightarrow{\text{difference}}$ Expected.

↓ crazy ↑

∴ Sex, Survived are related.

$$\text{Chi-squared Statistic} = \frac{(109-355)^2}{355} + \ldots$$

Disadvantages :

1. Works on categorical data
2. Independence of observation.
3. Sufficient sample size required.
4. No-variable interactions.

Mutual Information :

measure of the dependency b/w the two variables. It quantifies the amount of information obtained about one random variable through observing the other variable.

$\boxed{f_1 \quad f_2}$

Looking at f_1 how much I can tell about f_2.

$$M.I = \sum_{x \in X} \sum_{y \in Y} P(x,y) \log \left[\frac{P(x,y)}{P(x)\, P(y)} \right]$$

$P(x,y) \longrightarrow$ Joint prob. of x & y

$P(x) \longrightarrow$ marginal prob. of x

$P(y) \longrightarrow$ marginal prob. of y.

Sex	Survived
M	0
F	1
M	0
F	0
M	1

$\longrightarrow$

	0	1	
M	1: 2/5	2: 1/5	3/5
F	3: 1/5	4: 0/5	2/5
	3/5	2/5	

Joint prob.

Marginal Prob.

$$M.I = \frac{2}{5}\log\left(\frac{2/5}{3/5 \times 3/5}\right) + \frac{1}{5}\log\left(\frac{1/5}{3/5 \times 2/5}\right) + \frac{1}{5}\log\left(\frac{1/5}{3/5 \times 2/5}\right) + \frac{1}{5}\log\left(\frac{1/5}{2/5 \times 2/5}\right)$$

M.I $\longrightarrow$ sex & survived.

M.I ↑ if $P(x, y)$ ↑

M.I ↓ if $P(x)$, $P(y)$ ↓

Properties of M.I :

1. Non-negative.
2. Symmetric.
3. It can capture any kind of statistical dependency.

It can be also applied to numerical data.

Age, Survive $\longrightarrow$ Histogram $\longrightarrow$ Age is replaced by bin

eg:)

Age	Survive
35	0
60	1
52	1
16	1
27	0

Bins	0-20	21-50	51-80
Category $\longrightarrow$	#1	#2	#3

Disadvantages :

- Estimation difficulty.
- Assume large sample size
- Computation complexity.
- Difficulty with continuous vars.
- No direct indication of the nature of relationship.
- Doesn't account for Redundancy.

Feature selection | wrapper method

What is wrapper method?

Wrapper method involve using predictive model to score the combination of features.

$f_1, f_2, f_3, \ldots\ldots f_n \quad y$

subsets generated

$(f_1, f_2, y) \quad (f_2, f_3, y)$

ML algo   ML Algo

R^2 score R^2 score

Select highest R^2 score subset.

Steps :-

1. Subset Generation
2. Subset Evaluation.
3. Stopping Criteria.

Techniques in wrapper method :

1. Exhaustive feature selection.
2. forward selection
3. backward elemination.
4. Recursive feature elemination.

Exhaustive Feature Selection :

f_1 f_2 f_3 f_4 f_5 y.

↓ $(f_1 f_2 y)$ (f_1, f_3, y) ... $2^n - 1$ subsets.

Try all the subset

↓

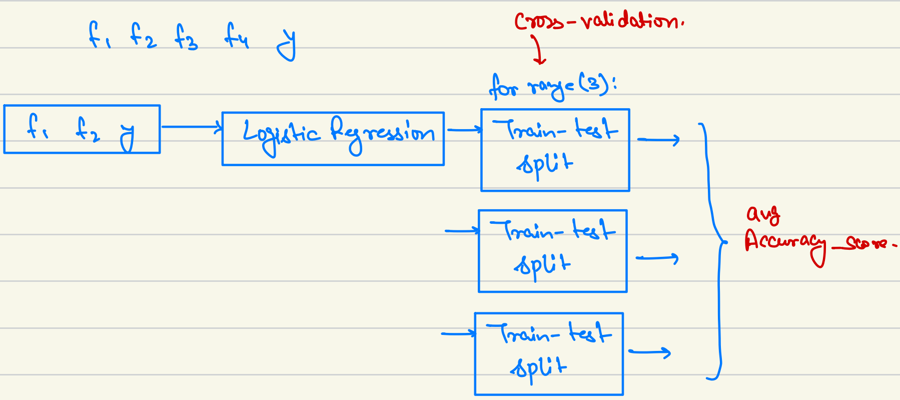
Select the best feature subset.

Disadvantage :

1. Computation complexity : Total no. of subset $(2^n - 1)$. Exponential in nature.
2. Cannot train big models (like with 10+ columns), Good till 10 columns.
3. Risk of overfitting
4. Requires a Good Evaluation Matric.

Cross validation :

f_1 f_2 f_3 f_4 y



Sequential Backward Elimination :

Also called backward elemination.

Accuracy

iter 1 $\boxed{f_1}$ $\boxed{f_2}$ $\boxed{f_3}$ $\boxed{f_4}$ $\longrightarrow$ model $\longrightarrow$ 0.89

iter 2 f_1 $\boxed{f_2}$ $\boxed{f_3}$ $\boxed{f_4}$ $\longrightarrow$ model $\longrightarrow$ 0.87

$\boxed{f_1}$ f_2 $\boxed{f_3}$ $\boxed{f_4}$ $\longrightarrow$ model $\longrightarrow$ 0.71

$\boxed{f_1}$ $\boxed{f_2}$ f_3 $\boxed{f_4}$ $\longrightarrow$ model $\longrightarrow$ 0.91 Best result.

$\boxed{f_1}$ $\boxed{f_2}$ $\boxed{f_3}$ f_4 $\longrightarrow$ model $\longrightarrow$ 0.65

Accuracy

iter 3 f_1 $\boxed{f_2}$ $\boxed{f_4}$ $\longrightarrow$ model $\longrightarrow$ 0.79

$\boxed{f_1}$ f_2 $\boxed{f_4}$ $\longrightarrow$ model $\longrightarrow$ 0.81

$\boxed{f_1}$ $\boxed{f_2}$ f_4 $\longrightarrow$ model $\longrightarrow$ 0.83

iter 4 f_1 $\boxed{f_2}$ $\longrightarrow$ model $\longrightarrow$ 0.63

$\boxed{f_1}$ f_2 $\longrightarrow$ model $\longrightarrow$ 0.53

$f_1 \quad f_2 \quad f_3 \quad f_4 \longrightarrow 0.88$

$\boxed{f_1 \quad f_2 \quad f_4 \longrightarrow 0.91}$ Best feature subset.

$f_1 \quad f_2 \longrightarrow 0.83$

$f_2 \longrightarrow 0.63$

$(f_1, f_2, f_4) \longrightarrow$ model train $\longrightarrow$ result.

Advantage → Faster approach.

Total models :

EFS → $2^n - 1$ → $O(2^n)$ exponential.

B.E → $\frac{n(n+1)}{2}$ → $O(n^2)$

Disadvantage → • Taking only best accuracy subset from every iter.
Leading to Local Selection.
We can miss the best subset.

Sequential Forward Selection :

f_1 f_2 f_3 f_4

Starts with 0 → y-mean

iter 1

f_1 → model → 0.63 select the best.

f_2 → model → 0.51

f_3 → model → 0.43

f_4 → model → 0.49

iter 2

f_1 f_2 → model → 0.63

f_1 f_3 → model → 0.71

f_1 f_4 → model → 0.80

iter 3

f_1 f_4 f_2 → model → 0.81

f_1 f_4 f_3 → model → 0.85

iter 4 f_1 f_2 f_3 f_4 → model → 0.83

f_1 → 0.63

f_1 f_4 → 0.80

f_1 f_4 f_3 → 0.85 Best for selection

f_1 f_4 f_3 f_2 → 0.83

$(f_1, f_4, f_3) \longrightarrow$ model $\longrightarrow$ result

Advantage $\rightarrow$ Faster approach.

Total models :

EFS $\rightarrow 2^n - 1 \longrightarrow O(2^n)$ exponential.

B.E $\rightarrow \frac{n(n+1)}{2} \rightarrow O(n^2)$

F.S $\rightarrow \frac{n(n+2)}{2} \longrightarrow O(n^2)$

How to select ? (if col = 100)

→ We need 10 : FS

→ We need 90 : B.E

Disadvantage → • Taking only best accuracy subset from every iter.
Leading to Local Selection.
We can miss the best subset.

Advantages of Wrapper Methods :

- Accuracy
- Interaction of features.

Disadvantages :

- Computational Complexity
- Risk of Overfitting.
- Model specific.

Feature selection| embedded method

Filter
↓
Feature interaction
(x_1, y)
(x_2, y)

Wrapper
↓
Computationally expensive.

Embedded ⟶ feature interaction + fast.

In embedded methods :

Data ⟶ Model ⟶ Model Training + feature Importance

Feature selection is embedded in ML model building.

Embedded Techniques :

ML model

- Coef_
  - Linear regression
  - Logistic regression
  - Ridge
  - Lasso
  - ElasticNet
- feature Importance
  - Decision Tree
  - Random forest
  - Gradient Boosting.

Linear Regression :

cgpa | iq | lpa

$$lpa = \beta_0 + \beta_1 \, cgpa + \beta_2 \, iq$$

Coef_ Coef_

Coef_

↳ Feature Importance

Assumptions :

1. Linearity
2. Independence
3. Homoscadesticity.
4. Normality.
5. No Multicollinearity.

Regularized Model :

It's linear regression with a penalty term in loss fn during training.

It prevents overfitting.

1. Ridge regression.
2. Lasso regression (Generally used for feature selection)
3. ElasticNET regression

HYBRID | WRAPPER.

Recursive Feature Elemination:

f_1 f_2 f_3 f_4 y

f_1 f_2 f_3 f_4 $\longrightarrow$ Model with coef_ | feature_importance_

$\downarrow$

Remove col. with least feature_importance_